

Crop Classification - Assessment

1 Build an image classification model using **TensorFlow or PyTorch** to classify:

Sunflower | Paddy | Wheat | Sugarcane | Maize

Tasks

1. Dataset Preparation

- Collect and organize images into the **5 classes**.
- Preprocess images: **resize, normalize, augment**.
- Split into **train / validation / test**.

2. Model

- Select a suitable CNN/deep-learning model.
- Explain your model choice.

3. Training

- Train the model and report key parameters such as epochs, batch size, optimizer, and learning rate.

4. Evaluation

- Report **accuracy, precision, recall, F1-score**.
- Provide a **confusion matrix**.
- Show training/validation accuracy and loss.

5. Analysis

- Identify difficult/confused classes.
- Mention any overfitting.
- Suggest how you would improve the model.

2 : Create a simple prediction script that takes an image and returns the crop class and confidence.

Dataset Sources

- <https://www.kaggle.com/datasets/mdwaquarazam/agricultural-crops-image-classification>
- <https://www.kaggle.com/datasets/imspash/flowers-dataset>
- <https://www.kaggle.com/datasets/vbookshelf/global-wheat-head-dataset-2021>
- <https://www.kaggle.com/datasets/aman2000jaiswal/agriculture-crop-images>

Other publicly available image datasets may also be used.